MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ENCODER, FAULT ISOLATION, KEY CODE MFG

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic fault isolation encoder module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:55Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.11 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MFG.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 80 contact pins positioned A1 through A40 and B1 through B40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/344 -1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8. End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

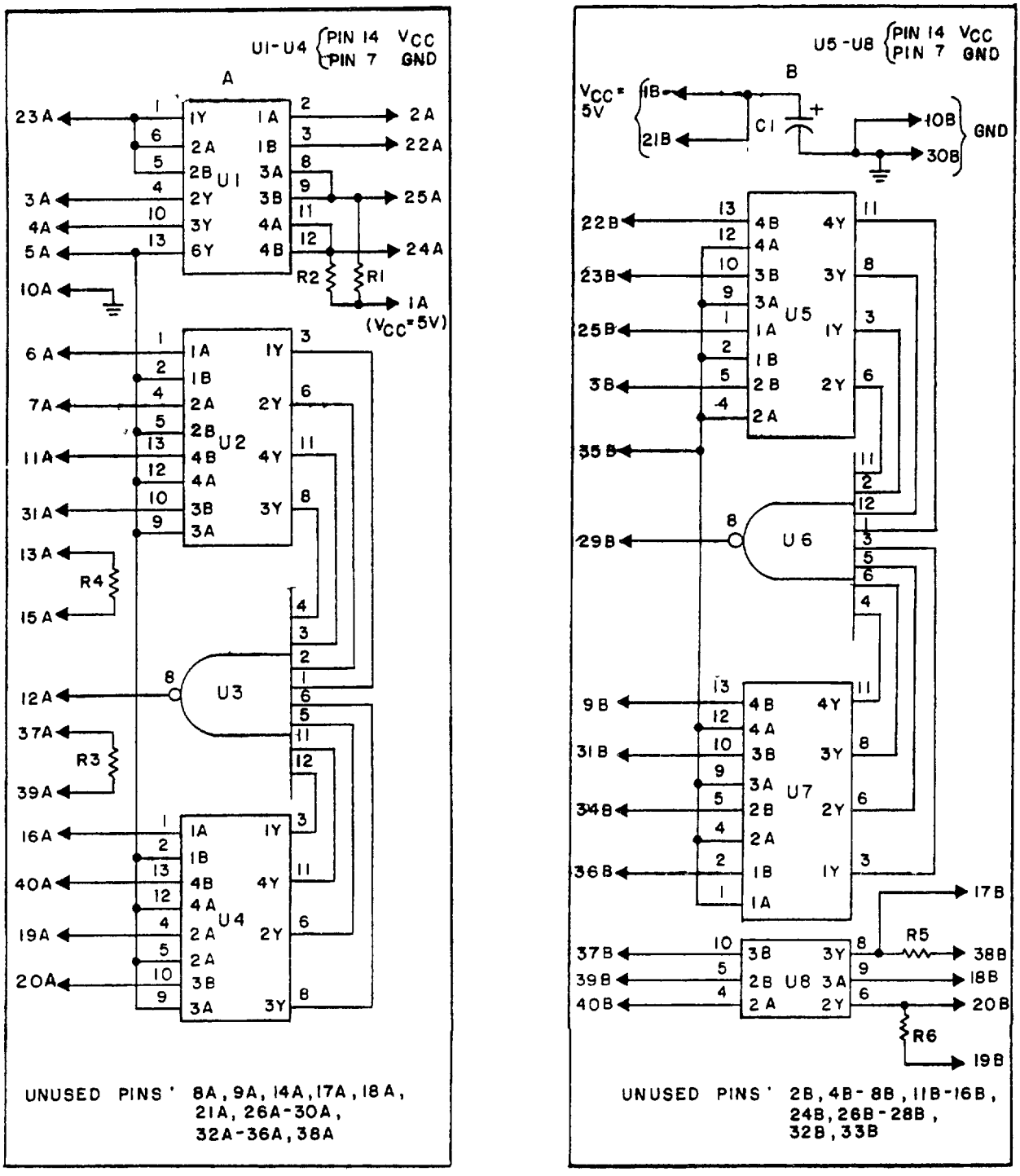
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.



NOTE: The following interconnections may be required
4A - 35B
2A - 12A
22A - 29B
23A - 40B
3A - 18B

FIGURE 1. MFG function diagram, fault isolation encoder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
A3-A16	Isolation	II	Q	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B4	Resistance	III	X	2024	2376	1914	2486	2024	2376	1914	2486	Ω
C1,C2	Resistance	III	X	138.0	162.0	130.5	169.5	138.0	162.0	130.5	169.5	Ω
D1,D2,D5, D6,D10,D11	NOR gate operation	IV	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
D3,D4, D7-D9,D12	NOR gate operation	IV	X	0	400	0	400	0	400	0	400	mV
E1,E11	Exclusive OR/8 INPUT NAND gate operation (module 1A)	V	X	0	400	0	400	0	400	0	400	mV
E2-E10, E12-E19	Exclusive OR/8 INPUT NAND gate operation (module 1A)	V	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
F1,F11	Exclusive OR/8 INPUT NAND gate operation (module 1B)	VI	X	0	400	0	400	0	400	0	400	mV
F2-F10, F12-F19	Exclusive OR/8 INPUT NAND gate operation (module 1B)	VI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
G1-G3,G5-G7	Independent NAND gate operation	VII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
G4,G8	Independent NAND gate operation	VII	X	0	400	0	400	0	400	0	400	mV
H1-H5, H11-H13,H18	Encoder fault isolation	VIII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
H6-H10, H14-H17	Encoder fault isolation	VIII	X	0	400	0	400	0	400	0	400	mV
J	Filter capacitance	IX	S	8	12	7	13	7	13	6	14	μF
K	Power supply current (V_{CC} max)	X	X	10	80	10	80	10	80	10	80	mA

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L1-L22	Input clamp diode voltage	XI	S	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
M1-M16	Input breakdown current	XII	S	0	200	0	200	0	200	0	200	µA
M17,M18	Input breakdown current	XII	S	0	100	0	100	0	100	0	100	µA
M19-M22	Input breakdown current	XII	S	0	1	0	1	0	1	0	1	mA
N1,N2	Output short circuit current	XIII	X	15	130	15	130	15	130	15	130	mA
N3-N6	Output short circuit current	XIII	X	30	130	30	130	30	130	30	130	mA
P1,P2	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	5	75	5	75	5	75	5	75	ns
P3,P4	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	10	150	10	150	10	150	10	150	ns
P5-P20	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	6	50	6	50	6	50	6	50	ns
P21-P24	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	5	25	5	25	5	25	5	25	ns
P25,P26	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	4	30	4	30	4	30	4	30	ns
P27,P28	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	8	60	8	60	8	60	8	60	ns
P29-P44	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	6	50	6	50	6	50	6	50	ns
P45-P48	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	4	40	4	40	4	40	4	40	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:55Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through P shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence. If during quality assurance testing test conditions A or B are performed, they shall be followed by the performance of all the tests in table I which are marked with an "X" in the test type column.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through P in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through P shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be nonoperating. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2 Following the life test and while still at the maximum class temperature, electrical tests B through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through P shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Power dissipation	400 mW maximum

6.4 Module function. This module contains two independent encoder networks and a voting circuit. The encoder networks each test eight inputs for either all high or all low values, determined by the state of the "detect 1's" input. If the "detect 1's" input is high, there will be a low output when all inputs are high If the "detect 1's" input is low, there will be a low output when all eight inputs are low.

The voting circuit operates with both inputs in the low state, when in this state the PASS output goes low.

Both 'PASS' and 'FAIL' outputs have a secondary output (via a 150 ohm resistor) for use with LED indicators, and both outputs are independently enabled from an external source.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:55Z
Check the source to verify that this is the current version before use.

6.5 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed.

a. Device pins should be in contact with conductive material except when being tested or in actual operation, to avoid build-up of static charge.

b. Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply (V_{CC}) is off.

d. All input signals and conditioning voltages should be removed before V_{CC} is disconnected.

e. Input voltages greater than the supply (V_{CC}) should not be applied as this will forward bias the protection input diodes.

f. All unused input leads should be connected to either 0 V REF(GND) or V_{CC} (device supply), whichever is appropriate for the logic circuit involved.

6.6 Caution. This caution applies to each test figure. Failure to comply with the following procedures may result in irreversable module damage.

a. All test equipment should be grounded.

b. V_{CC} shall be applied before applying input signals.

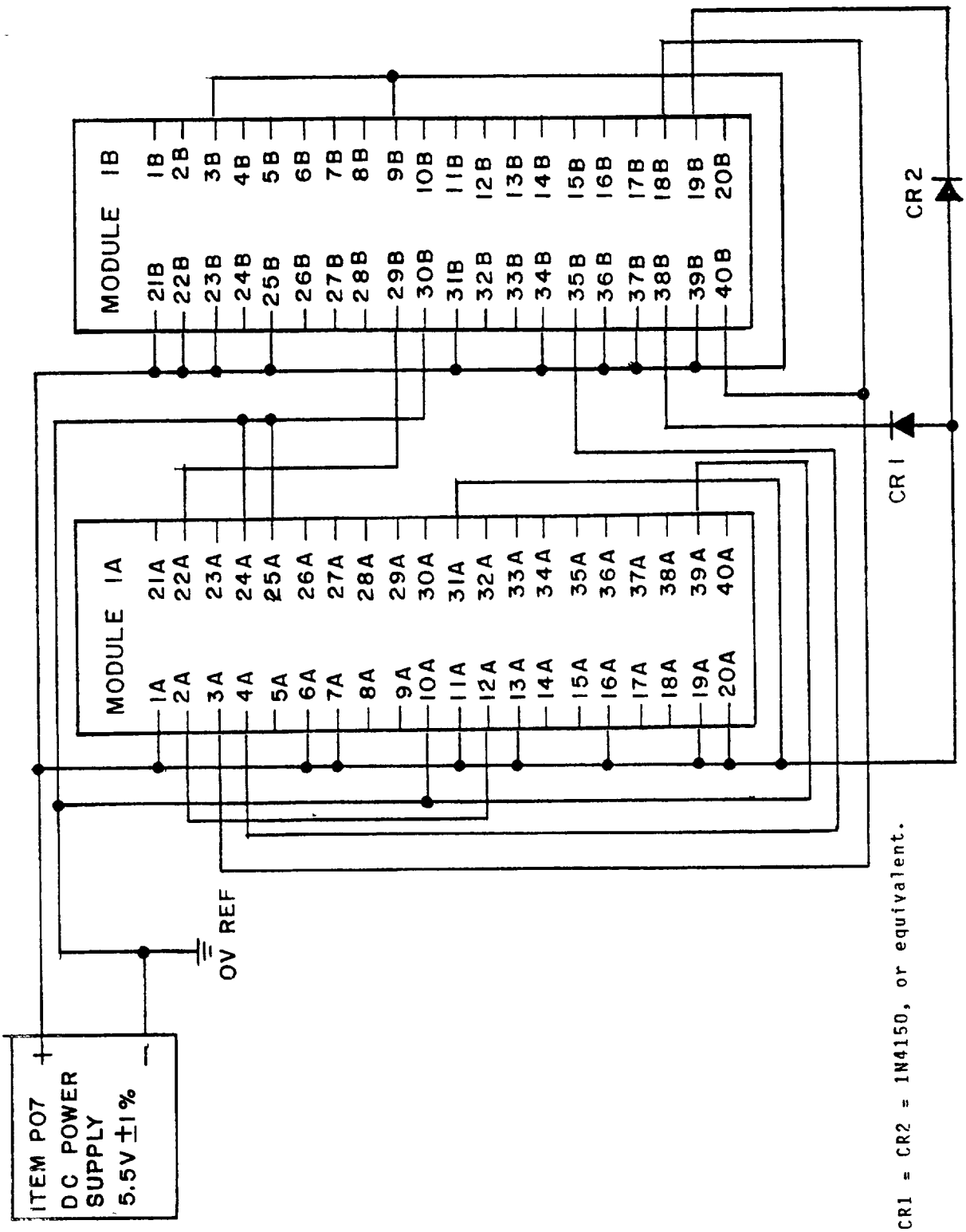
c. All inputs signals and conditioning voltages should be removed before V_{CC} is disconnected.

Agent:
DLA - ES

Preparing activity
Navy - EC

(Project 5963-N436)

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.



NOTE: CR1 = CR2 = 1N4150, or equivalent.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	Frame	All pins	10	∞	MΩ	3
A2	Isolation, used to unused pins	1A-7A,10A-13A, 15A,16A,19A, 20A,22A-25A, 31A,37A,39A, 40A,1B,3B,9B, 10B,17B-23B, 25B,29B-31B, 34B-40B	8A,9A,14A,17A, 18A,21A,26A-30A,32A-36A, 38A,2B,4B-8B, 11B-16B,24B, 26B-28B,32B, 33B	"	"	"	"
A3	Isolation, unused pins	8A	9A,14A,17A, 18A,21A,26A-30A,32A-36A, 38A	"	"	"	"
A4	"	9A	14A,17A,18A, 21A,26A-30A, 32A-36A,38A	"	"	"	4
A5	"	14A	17A,18A,21A, 26A-30A,32A-36A,38A	"	"	"	"
A6	"	17A	18A,21A,26A-30A,32A-36A, 38A	"	"	"	"
A7	"	18A	21A,26A-30A, 32A-36A,38A	"	"	"	"
A8	"	21A	26A-30A,32A-36A,38A	"	"	"	"
A9	"	26A-30A	32A-36A,38A	"	"	"	"
A10	"	32A-36A	38A	"	"	"	"
A11	"	2B	4B-8B,11B-16B,24B,26B-28B,32B,33B	"	"	"	"
A12	"	4B-8B	11B-16B,24B, 26B-28B,32B, 33B	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

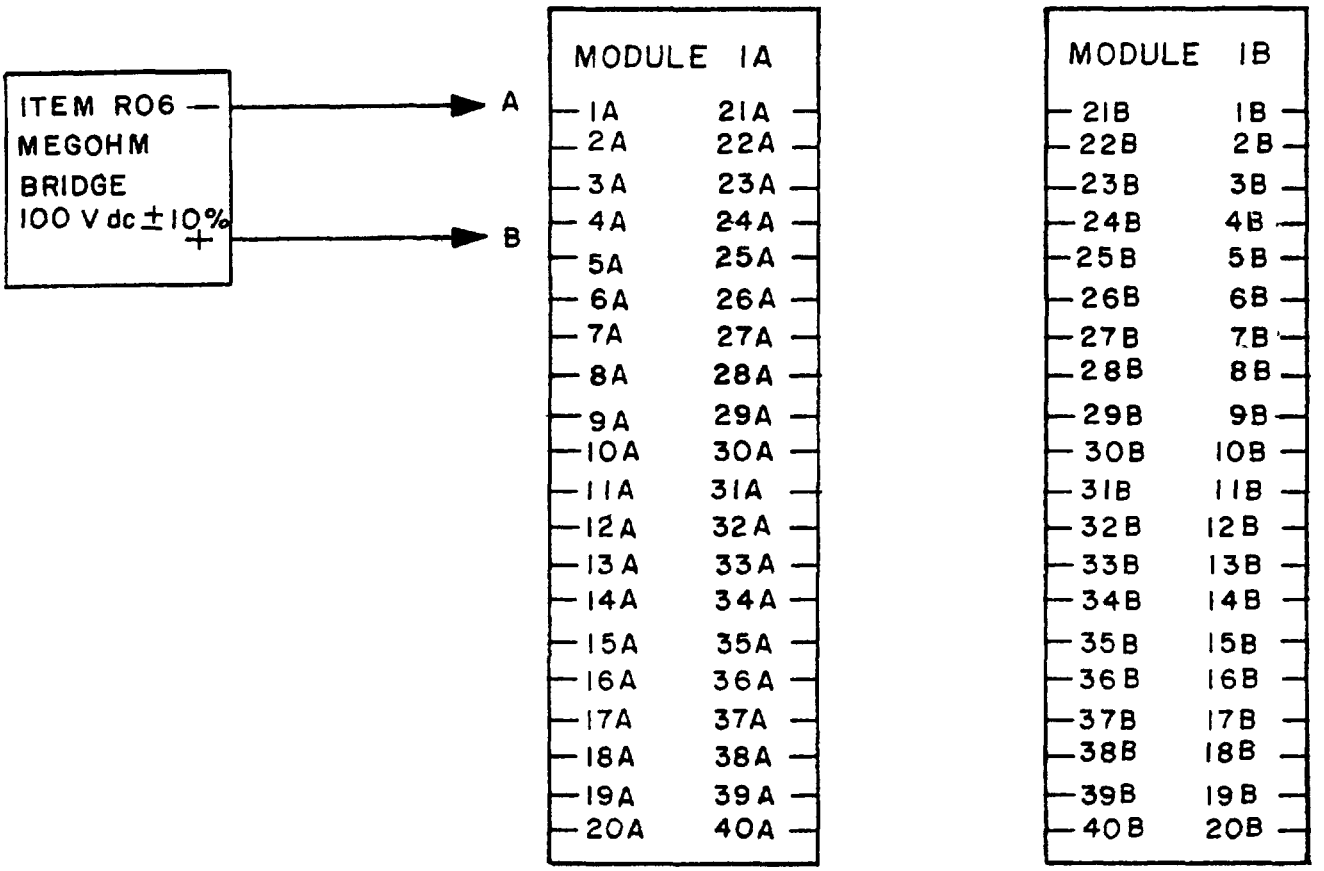
TABLE II. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A13	Isolation, unused pins	11B-16B	24B,26B-28B, 32B,33B	10	∞	MΩ	4
A14	"	24B	26B-28B,32B, 33B	"	"	"	"
A15	"	26B-28B	32B,33B	"	"	"	"
A16	"	32B	33B	"	"	"	"

TABLE III. Resistance test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	Resistance	15A	2024	2376	Ω	5
B2	"	24A	"	"	"	"
B3	"	25A	"	"	"	"
B4	"	39A	"	"	"	"
C1	"	17B	138.0	162.0	"	"
C2	"	19B	138.0	162.0	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.



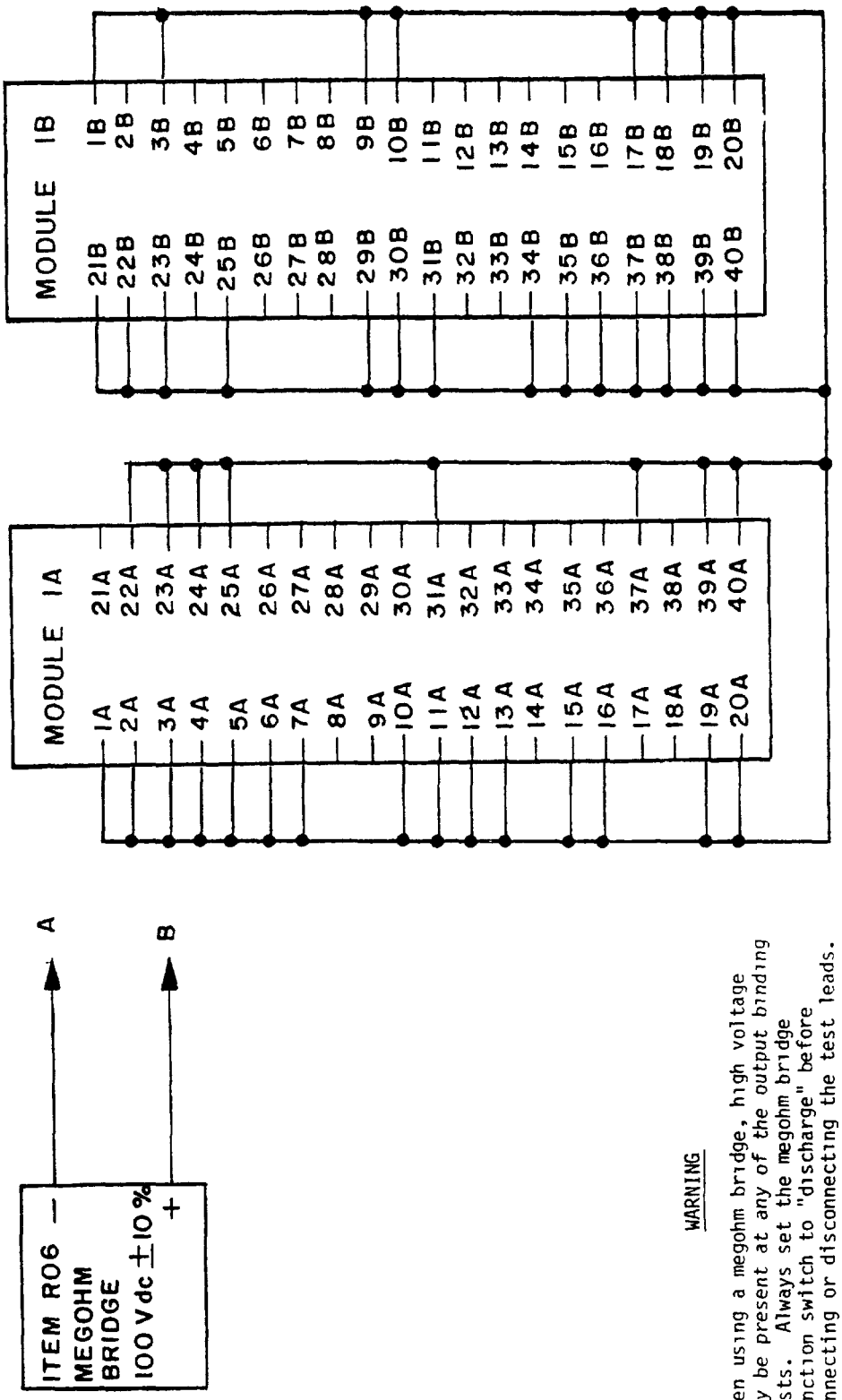
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 3. Isolation (frame and used to unused pins) test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:55Z
Check the source to verify that this is the currentversion before use.



WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 4. Isolation (unused pins) test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

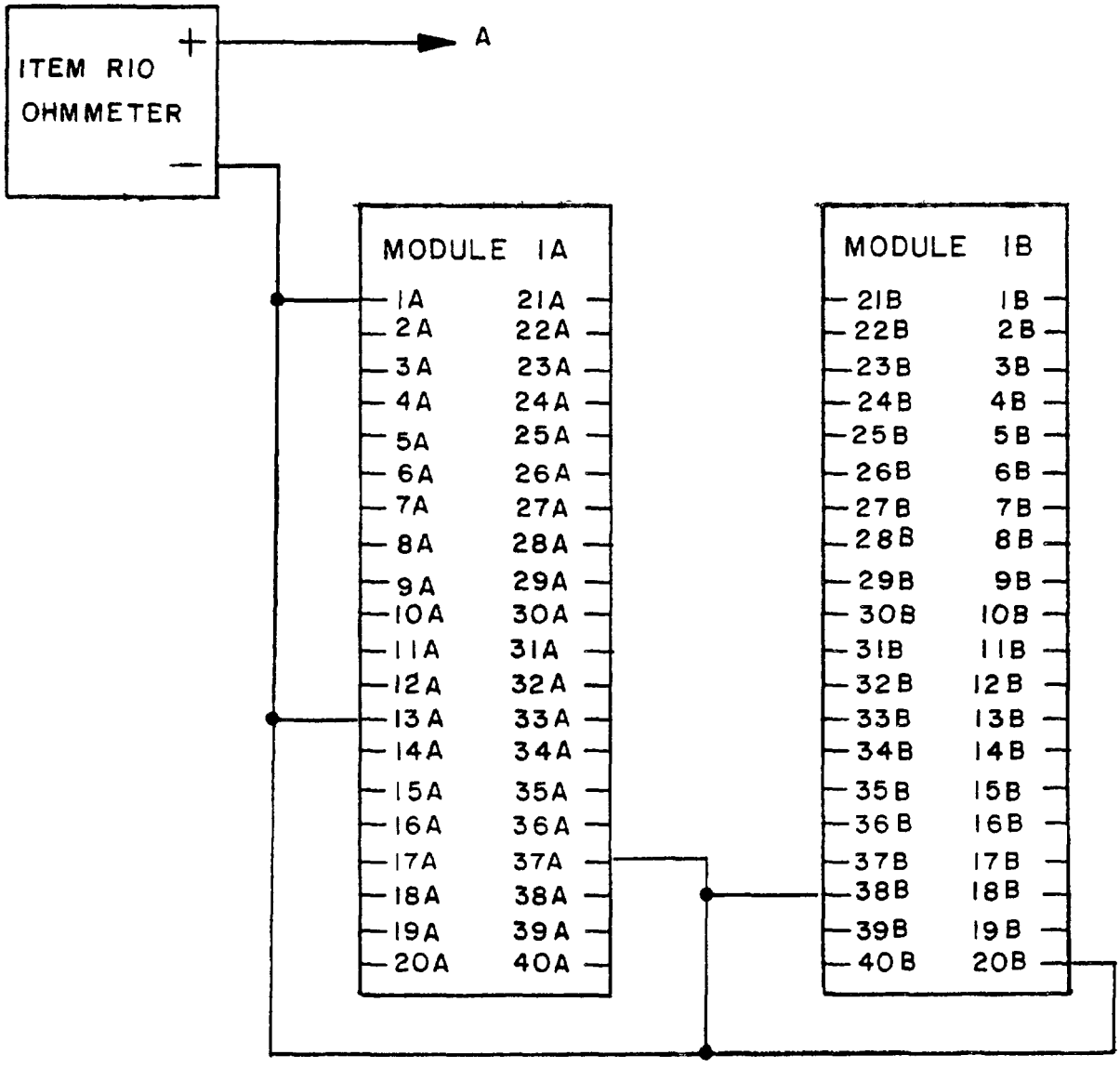


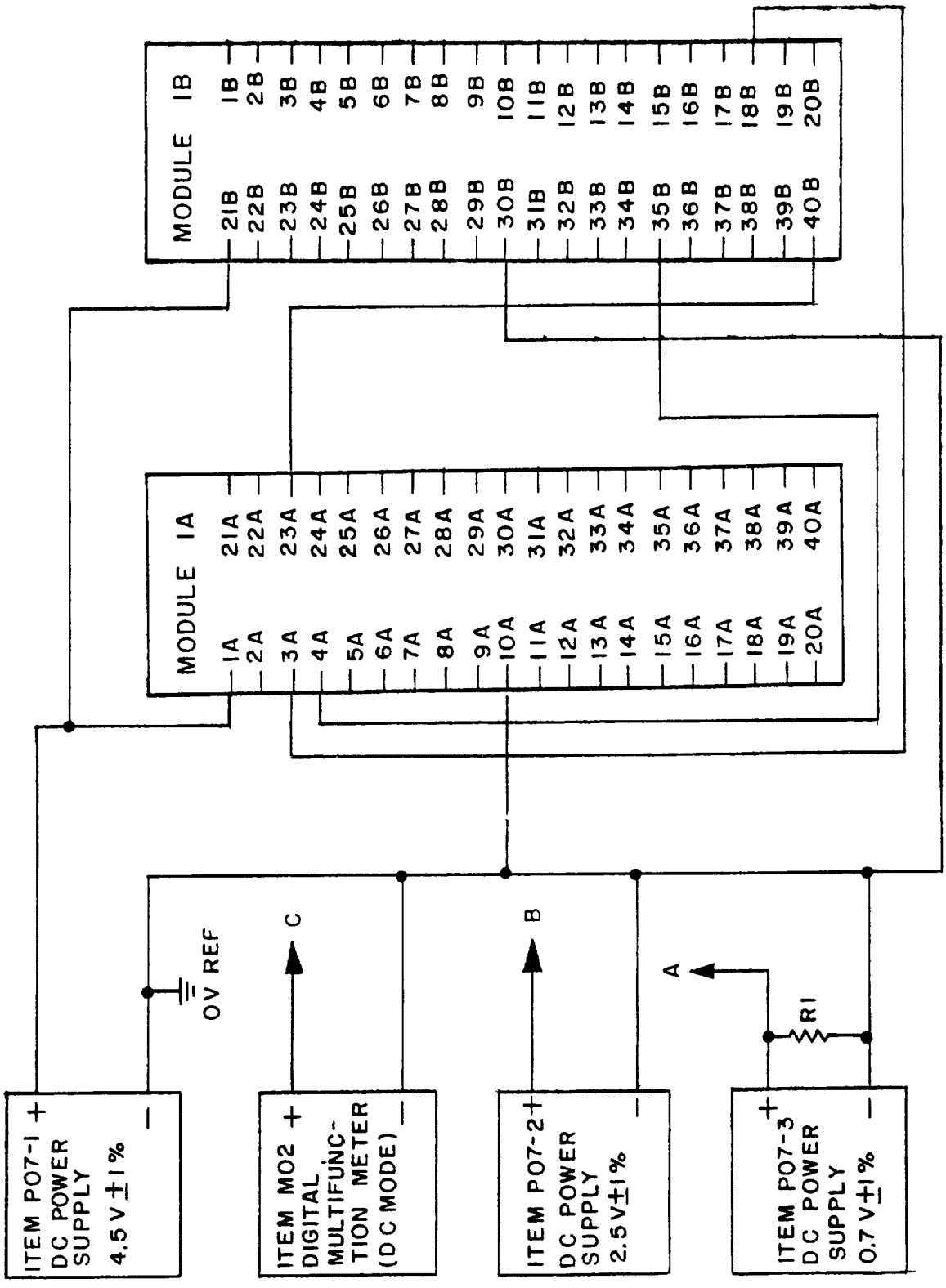
FIGURE 5. Resistance test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE IV. NOR gate operation test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	NOR gate operation	Open	Open	24A	2.5	4.5	V	6
D2	"	"	"	25A	2.5	4.5	V	"
D3	"	"	"	5A	0	400	mV	"
D4	"	"	"	4A	0	400	mV	"
D5	"	24A	"	5A	2.5	4.5	V	"
D6	"	25A	"	4A	2.5	4.5	V	"
D7	"	Open	2A,22A	23A	0	400	mV	"
D8	"	2A	22A	"	"	"	"	"
D9	"	22A	2A	"	"	"	"	"
D10	"	2A,22A	Open	"	2.5	4.5	V	"
D11	"	Open	2A,22A	3A	2.5	4.5	V	"
D12	"	2A,22A	Open	3A	0	400	mV	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.



NOTE. R1 = 475Ω ±5%, 1/8 W.

FIGURE 6. NOR gate operation test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the currentversion before use.

TABLE V. Exclusive OR/8 INPUT NAND gate operation (module 1A) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	Exclusive OR/8 INPUT NAND gate operation (module 1A)	6A,7A,11A, 16A,19A, 20A,31A,40A	Open	12A	0	400	mV	7
E2	"	6A,7A,11A, 16A,19A,20A, 31A,40A	24A	"	2.5	4.5	V	"
E3	"	7A,11A,16A, 19A,20A,31A, 40A	6A	"	"	"	"	"
E4	"	6A,11A,16A, 19A,20A,31A, 40A	7A	"	"	"	"	"
E5	"	6A,7A,16A, 19A,20A,31A, 40A	11A	"	"	"	"	"
E6	"	6A,7A,11A, 19A,20A,31A, 40A	16A	"	"	"	"	"
E7	"	6A,7A,11A, 16A,20A,31A, 40A	19A	"	"	"	"	"
E8	"	6A,7A,11A, 16A,19A,31A, 40A	20A	"	"	"	"	"
E9	"	6A,7A,11A, 16A,19A,20A, 40A	31A	"	"	"	"	"
E10	"	6A,7A,11A, 16A,19A,20A, 31A	40A	"	"	"	"	"
E11	"	Open	6A,7A,11A, 16A,19A, 20A,24A, 31A,40A	"	0	400	mV	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE V. Exclusive OR/8 INPUT NAND gate operation (module 1A) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E12	Exclusive OR/8 INPUT NAND gate operation (module 1A)	6A	7A,11A, 16A,19A, 20A,24A, 31A,40A	12A	2.5	4.5	V	7
E13	"	7A	6A,11A, 16A,19A, 20A,24A, 31A,40A	"	"	"	"	"
E14	"	11A	6A,7A, 16A,19A, 20A,24A, 31A,40A	"	"	"	"	"
E15	"	16A	6A,7A, 11A,19A, 20A,24A, 31A,40A	"	"	"	"	"
E16	"	19A	6A,7A, 11A,16A, 20A,24A, 31A,40A	"	"	"	"	"
E17	"	20A	6A,7A, 11A,16A, 19A,24A, 31A,40A	"	"	"	"	"
E18	"	31A	6A,7A, 11A,16A, 19A,20A, 24A,40A	"	"	"	"	"
E19	"	40A	6A,7A, 11A,16A, 19A,20A, 24A,31A	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE VI. Exclusive OR/8 INPUT NAND gate operation (module 1B) test.

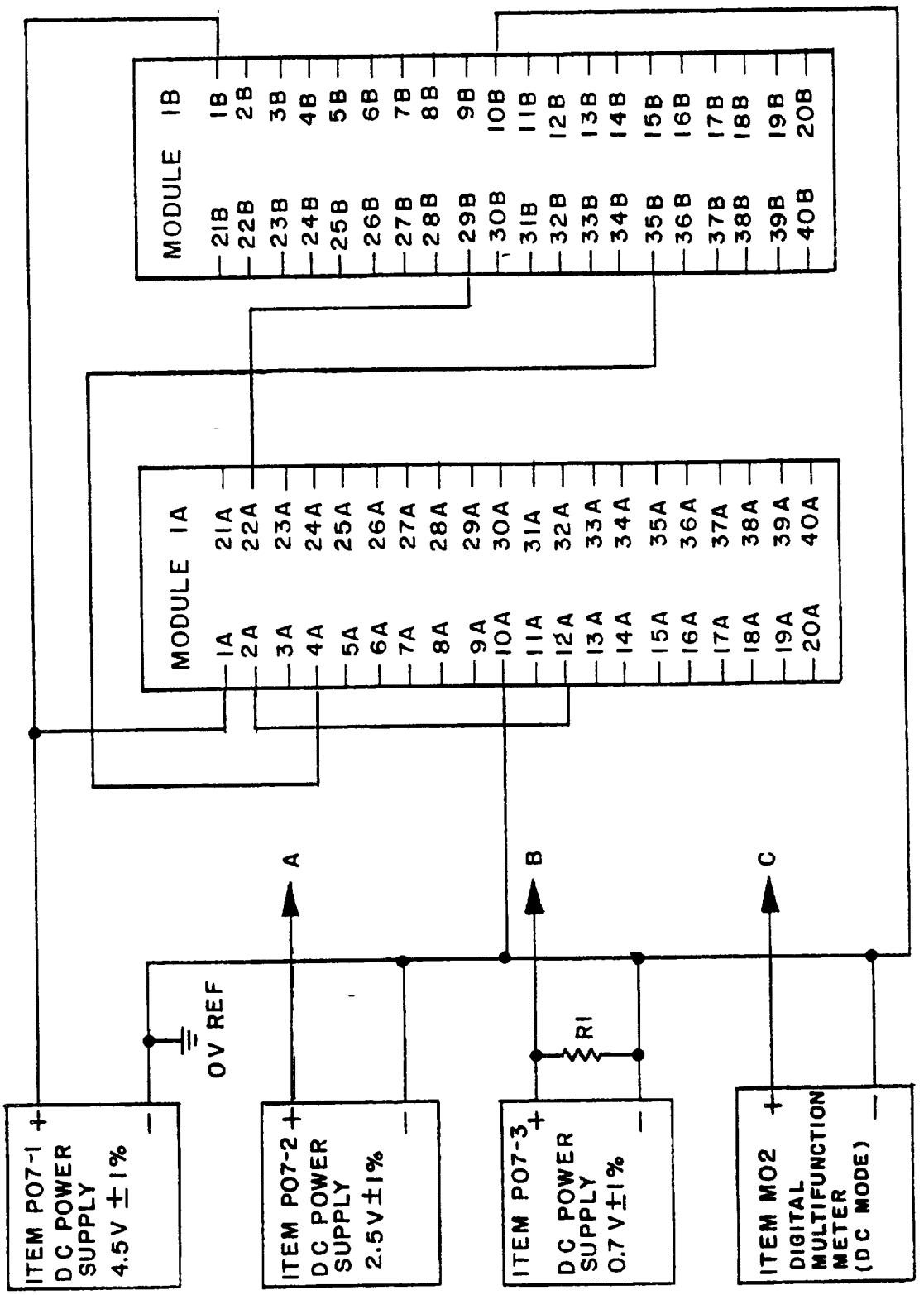
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	Exclusive OR/8 INPUT NAND gate operation (module 1B)	22B,23B,25B, 31B,34B,36B, 3B,9B	Open	29B	0	400	mV	7
F2	"	22B,23B,25B, 31B,34B,36B, 3B,9B	25A	"	2.5	4.5	V	"
F3	"	23B,25B,31B, 34B,36B,3B, 9B	22B	"	"	"	"	"
F4	"	22B,25B,31B, 34B,36B,3B, 9B	23B	"	"	"	"	"
F5	"	22B,23B,31B, 34B,36B,3B, 9B	25B	"	"	"	"	"
F6	"	22B,23B,25B, 34B,36B,3B, 9B	31B	"	"	"	"	"
F7	"	22B,23B,25B, 31B,36B,3B,9B	34B	"	"	"	"	"
F8	"	22B,23B,25B, 31B,34B,3B,9B	36B	"	"	"	"	"
F9	"	22B,23B,25B, 31B,34B,36B,9B	3B	"	"	"	"	"
F10	"	22B,23B,25B, 31B,34B,36B,3B	9B	"	"	"	"	"
F11	"	Open	25A,22B,23B, 25B,31B,34B, 36B,3B,9B	"	0	400	mV	"
F12	"	22B	25A,23B,25B, 31B,34B,36B, 3B,9B	"	2.5	4.5	V	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE VI. Exclusive OR/8 INPUT NAND gate operation test (module 1B) - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F13	Exclusive OR/8 INPUT NAND gate operation (module 1B)	23B	25A,22B,25B, 31B,34B,36B, 3B,9B	29B	2.5	4.5	V	7
F14	"	25B	25A,22B,23B, 31B,34B,36B, 3B,9B	"	"	"	"	"
F15	"	31B	25A,22B,23B, 25B,34B,36B, 3B,9B	"	"	"	"	"
F16	"	34B	25A,22B,23B, 25B,31B,36B, 3B,9B	"	"	"	"	"
F17	"	36B	25A,22B,23B, 25B,31B,34B, 3B,9B	"	"	"	"	"
F18	"	3B	25A,22B,23B, 25B,31B,34B, 36B,9B	"	"	"	"	"
F19	"	9B	25A,22B,23B, 25B,31B,34B, 36B,3B	"	"	"	"	"

Source: https://assist.dla.mil - Downloaded: 2026-02-20T17:57Z
Check the source to verify that this is the current version before use.



NOTE R1 = 50Ω ±10%, 1 W.

FIGURE 7. Exclusive OR/8 input NAND gate operation (modules 1A and 1B) for tables V and VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE VII. Independent NAND gate operation test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Independent NAND gate operation	Open	18B,37B	17B	2.5	4.5	V	8
G2	"	18B	37B	"	"	"	"	"
G3	"	37B	18B	"	"	"	"	"
G4	"	18B,37B	Open	"	0	400	mV	"
G5	"	39B	40B	20B	2.5	4.5	V	"
G6	"	40B	39B	"	"	"	"	"
G7	"	Open	39B,40B	"	"	"	"	"
G8	"	39B,40B	Open	"	0	400	mV	"

Source: http://esist.dla.mil — Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

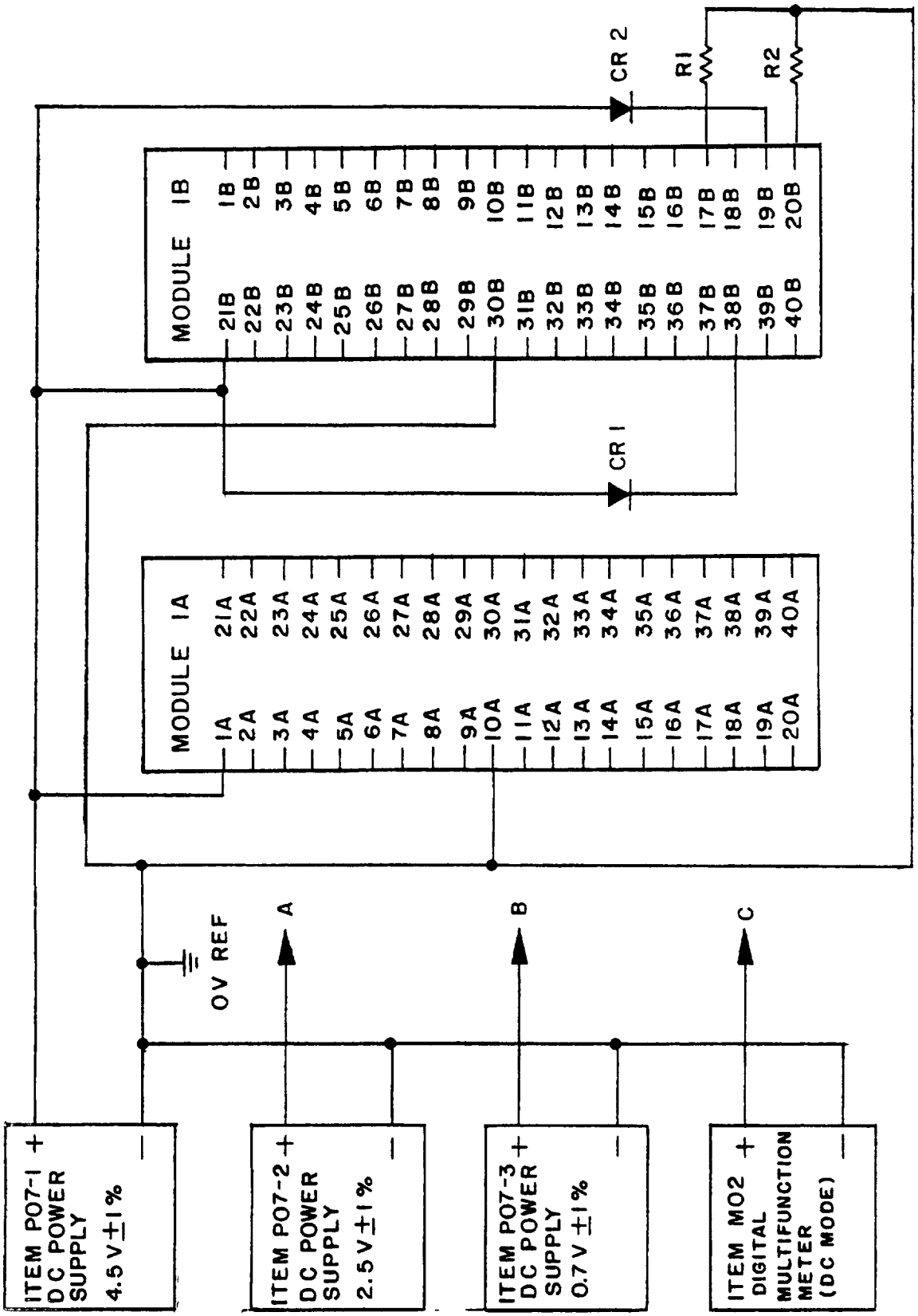


FIGURE 8. Independent NAND gate operation test circuit for table VII.

NOTES:
1. R1 = R2 = 2.08 kΩ ±1%, 1/8 W.
2. CR1 = CR2 = 1N4150, or equivalent.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE VIII. Encoder fault isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Encoder fault isolation	Open	17B	2.5	4.5	V	9
H2	"	Open	20B	"	"	"	"
H3	"	24A	"	"	"	"	"
H4	"	25A	"	"	"	"	"
H5	"	24A,25A	"	"	"	"	"
H6	"	Open	"	0	400	mV	"
H7	"	Open	17B	"	"	"	"
H8	"	24A	"	"	"	"	"
H9	"	25A	"	"	"	"	"
H10	"	24A,25A	"	"	"	"	"
H11	"	Open	20B	2.5	4.5	V	"
H12	"	24A	"	"	"	"	"
H13	"	25A	"	"	"	"	"
H14	"	24A,25A	"	0	400	mV	"
H15	"	Open	17B	"	"	"	"
H16	"	24A	"	"	"	"	"
H17	"	25A	"	"	"	"	"
H18	"	24A,25A	"	2.5	4.5	V	"

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:55Z
Check the source to verify that this is the current version before use.

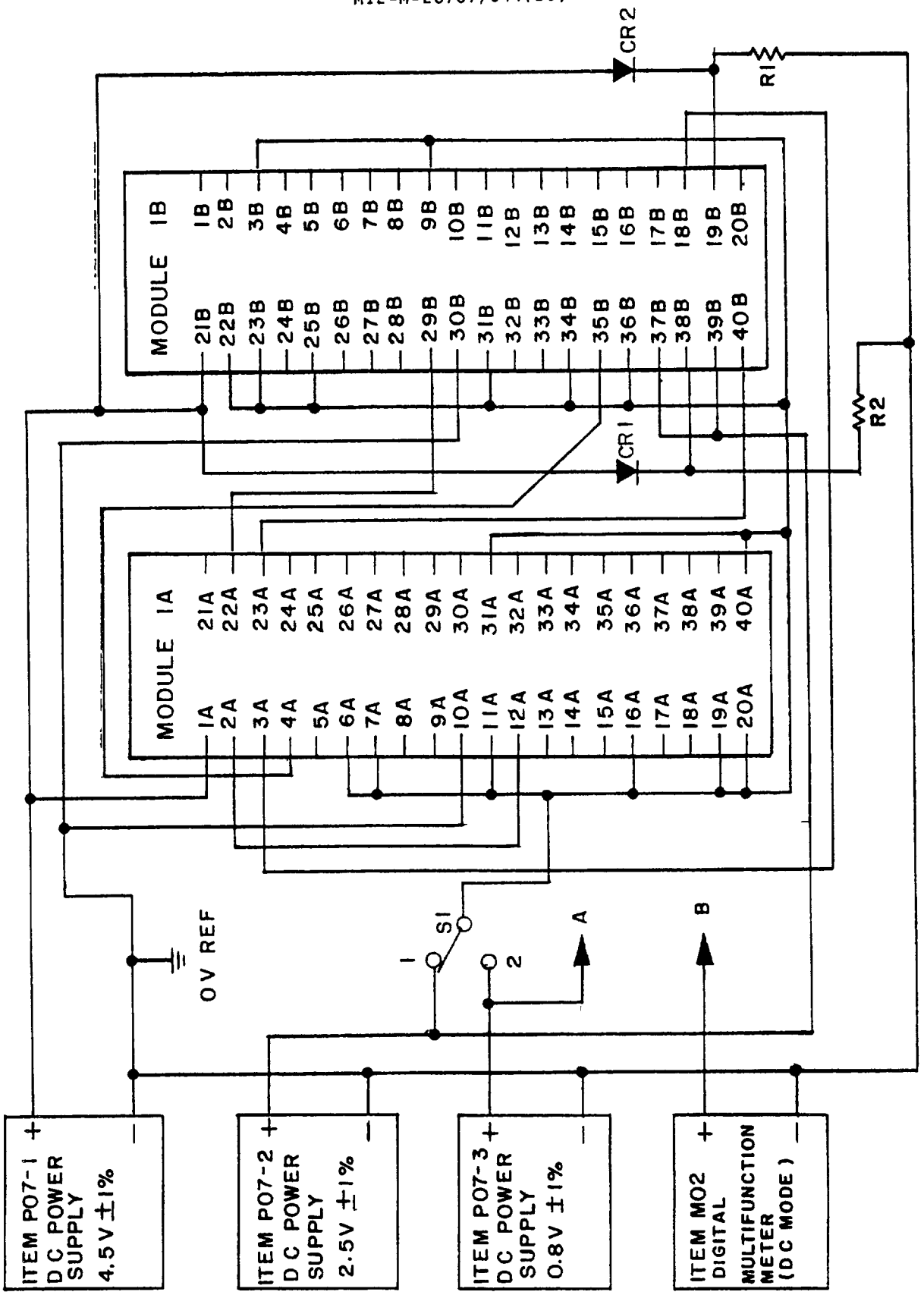


FIGURE 9. Encoder fault isolation test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:55Z
Check the source to verify that this is the currentversion before use.

NOTES:
1. R1 = R2 = 2.08 kΩ ±1%, 1/8 W.
2. S1 = SPDT switch.
3. CR1 = CR2 = 1N4150, or equivalent.
4. Set switch S1 to position 1 for test conditions H1, H3-H6, and H8-H10 and to position 2 for test conditions H2, H7, and H11-H18.

FIGURE 9. Encoder fault isolation test circuit for table VIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.

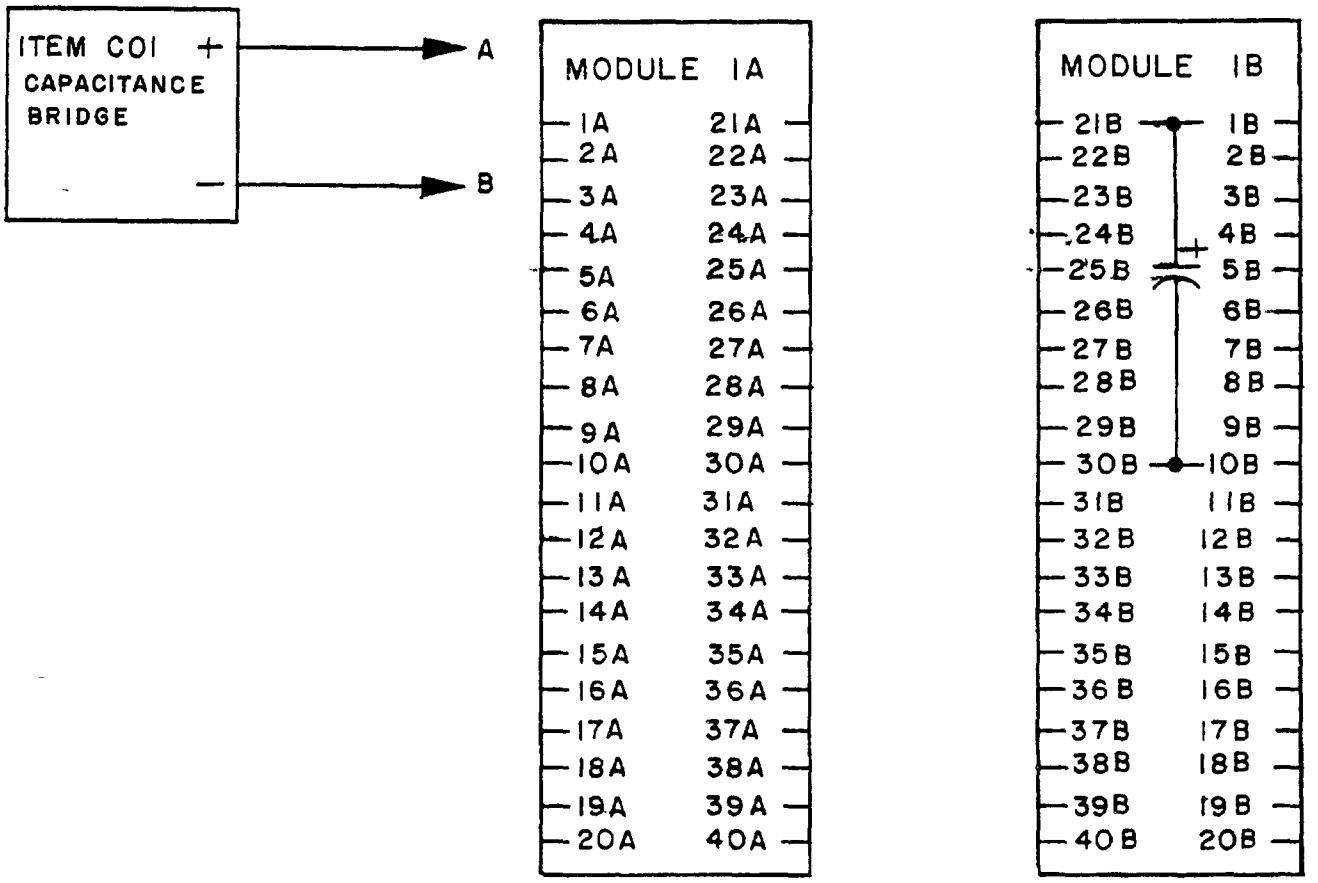
TABLE IX. Filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J	Filter capacitance	1B	10B	8	12	µF	10

TABLE X. Power supply current (V_{CC} max) test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
K	Power supply current (V_{CC} max)	10	80	mA	11

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:55Z
Check the source to verify that this is the current version before use.



NOTE: Item C01 setup:
Frequency = 1 kHz.
V_{pk} = 0.4 V maximum.

FIGURE 10. Filter capacitance test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.

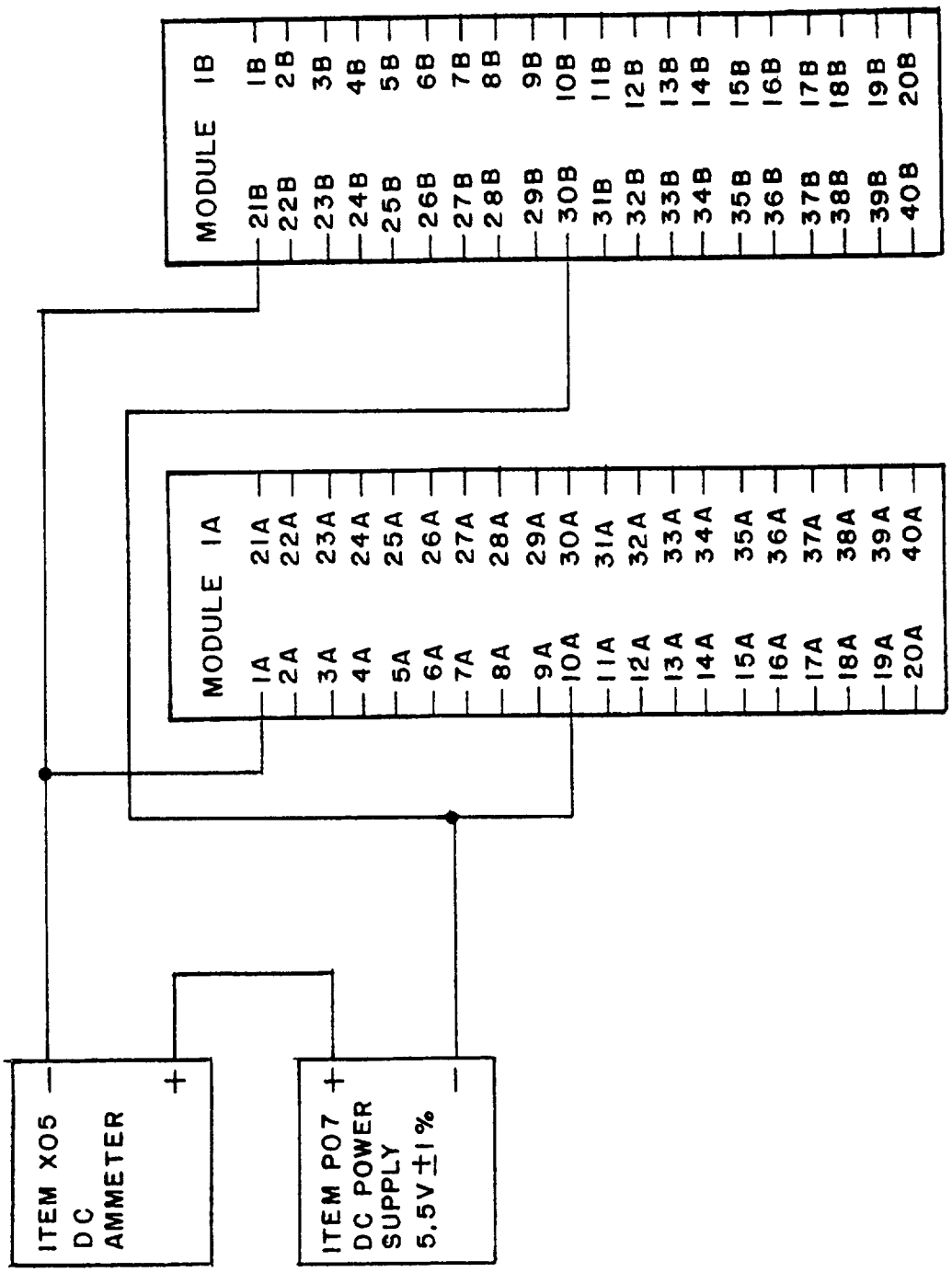


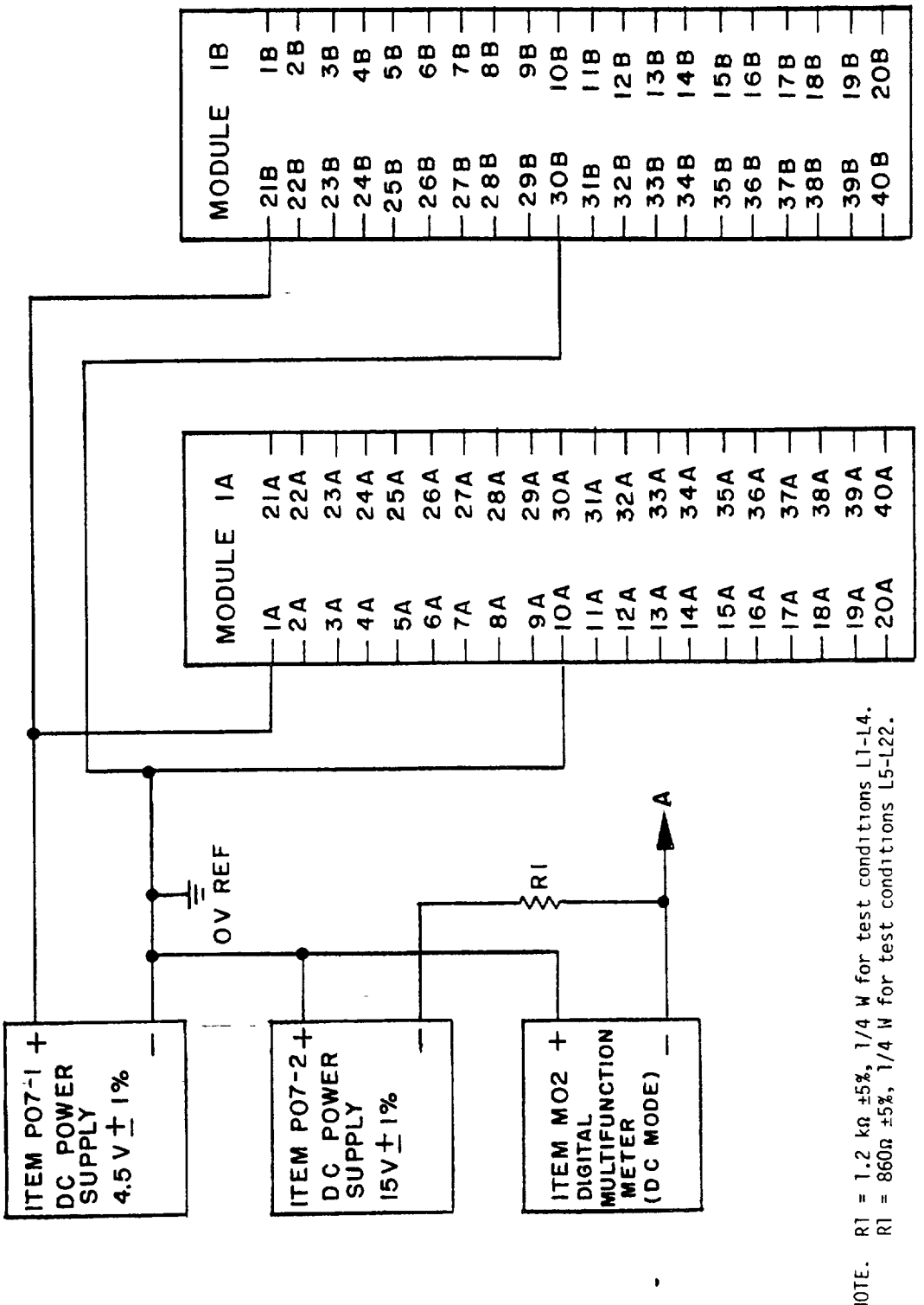
FIGURE 11. Power supply current (V_{CC} max) test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE XI. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Input clamp diode voltage	39B	0.4	1.5	V	12
L2	"	40B	"	"	"	"
L3	"	37B	"	"	"	"
L4	"	18B	"	"	"	"
L5	"	2A	"	"	"	"
L6	"	22A	"	"	"	"
L7	"	6A	"	"	"	"
L8	"	7A	"	"	"	"
L9	"	11A	"	"	"	"
L10	"	16A	"	"	"	"
L11	"	19A	"	"	"	"
L12	"	20A	"	"	"	"
L13	"	31A	"	"	"	"
L14	"	40A	"	"	"	"
L15	"	22B	"	"	"	"
L16	"	23B	"	"	"	"
L17	"	25B	"	"	"	"
L18	"	31B	"	"	"	"
L19	"	34B	"	"	"	"
L20	"	36B	"	"	"	"
L21	"	3B	"	"	"	"
L22	"	9B	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.



NOTE. R1 = 1.2 kΩ ±5%, 1/4 W for test conditions L1-L4.
R1 = 860Ω ±5%, 1/4 W for test conditions L5-L22.

FIGURE 12. Input clamp diode voltage test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE XII. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Input breakdown current	6A	Open	0	200	µA	13
M2	"	7A	"	"	"	"	"
M3	"	11A	"	"	"	"	"
M4	"	16A	"	"	"	"	"
M5	"	19A	"	"	"	"	"
M6	"	20A	"	"	"	"	"
M7	"	31A	"	"	"	"	"
M8	"	40A	"	"	"	"	"
M9	"	22B	"	"	"	"	"
M10	"	23B	"	"	"	"	"
M11	"	25B	"	"	"	"	"
M12	"	31B	"	"	"	"	"
M13	"	34B	"	"	"	"	"
M14	"	36B	"	"	"	"	"
M15	"	3B	"	"	"	"	"
M16	"	9B	"	"	"	"	"
M17	"	2A	22A	"	100	"	"
M18	"	22A	2A	"	100	"	"
M19	"	39B	40B	"	1	mA	"
M20	"	40B	39B	"	"	"	"
M21	"	37B	18B	"	"	"	"
M22	"	18B	37B	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

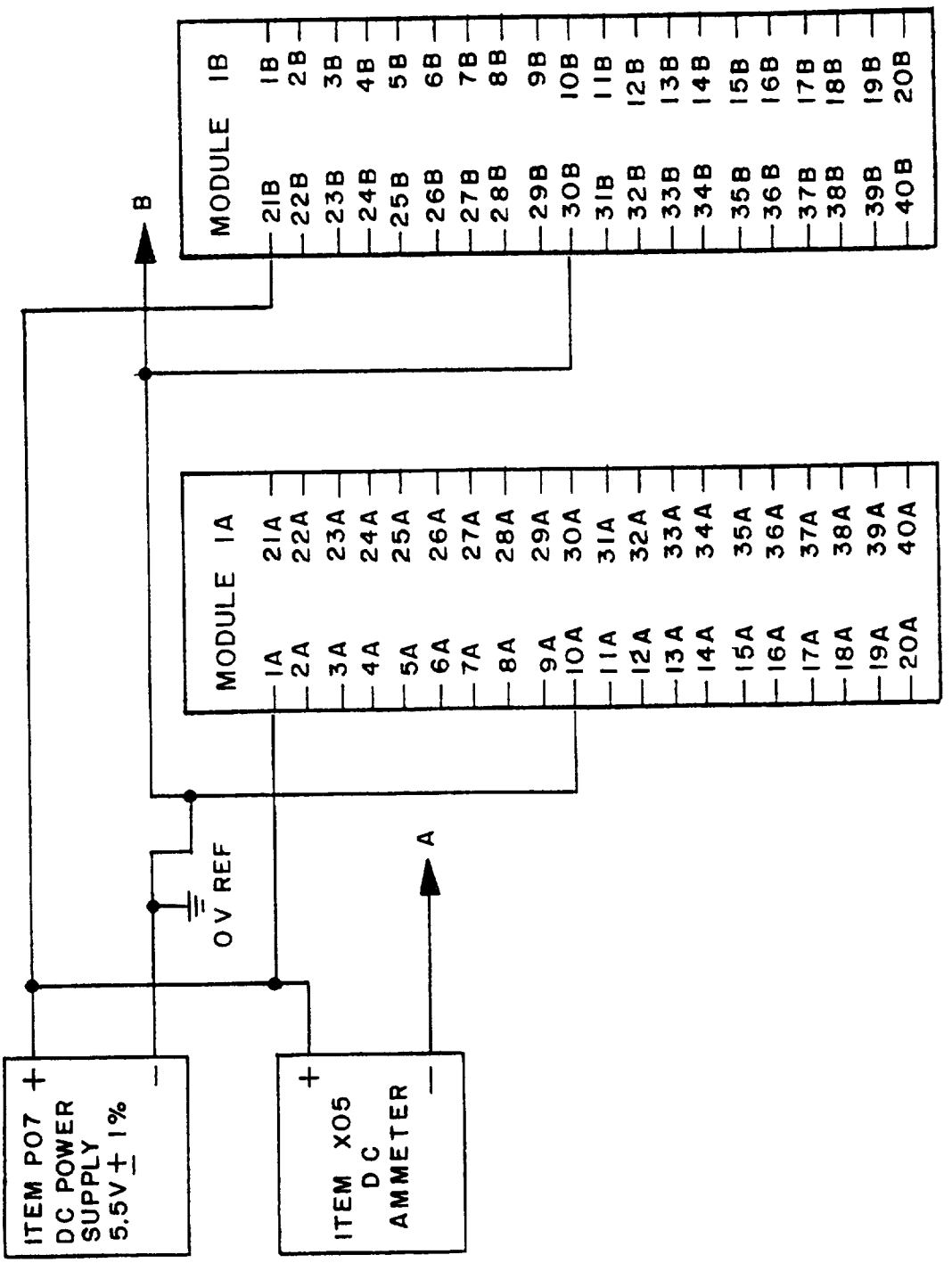


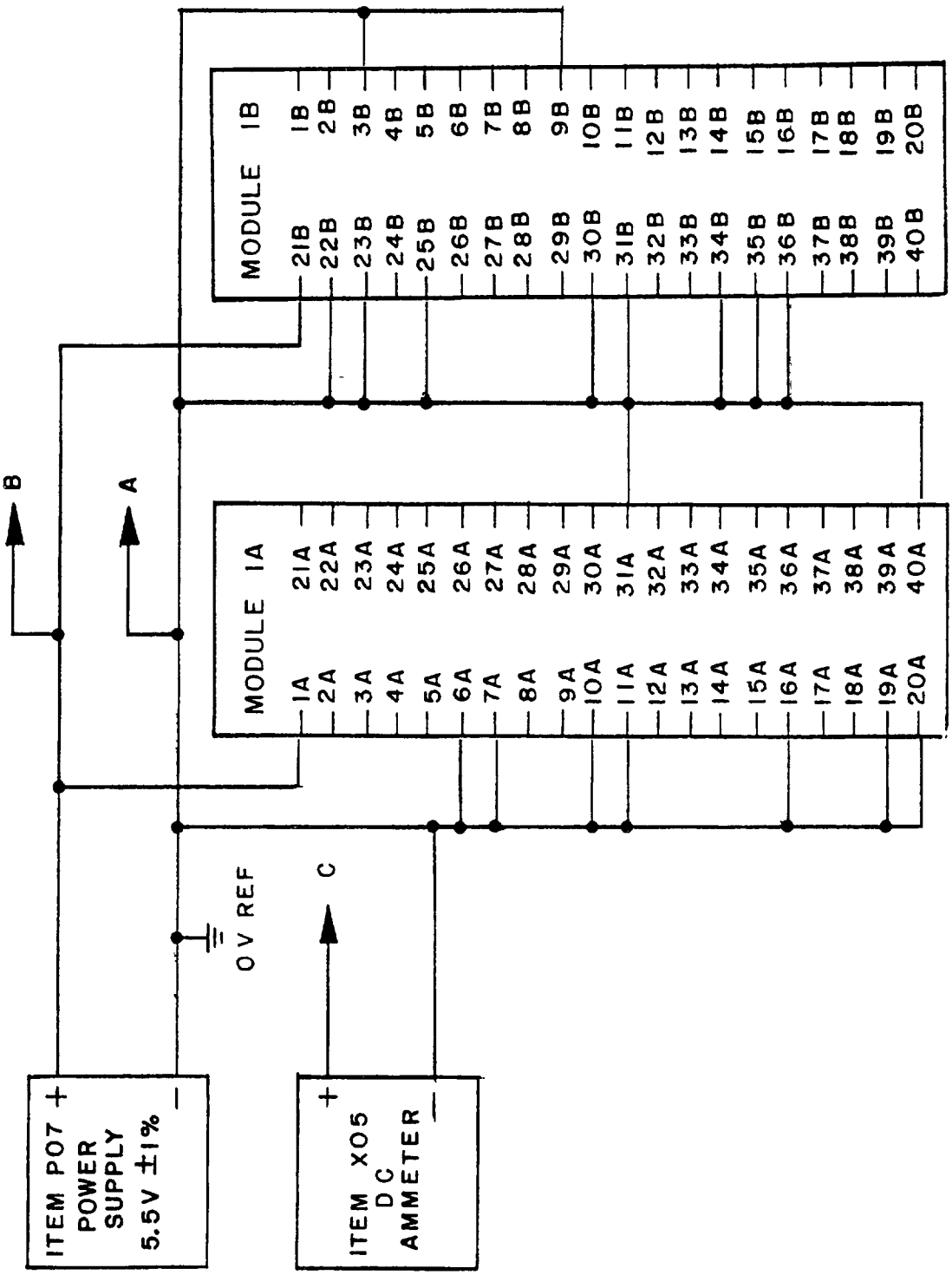
FIGURE 13. Input breakdown current test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE XIII. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Output short circuit current	Open	Open	12A	15	130	mA	14
N2	"	"	Open	29B	15	"	"	"
N3	"	"	2A,22A	3A	30	"	"	"
N4	"	2A,22A	Open	23A	"	"	"	"
N5	"	25A	"	4A	"	"	"	"
N6	"	24A	"	5A	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.



NOTES:
1. Duration of the test must not exceed five seconds.
2. Only one output to be shorted at one time.

FIGURE 14. Output short circuit current test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Propagation delay time (t_{PLH}), C_L = 50 pF	2A	22A	23A,40B	5	75	ns	15,16,18
P2	"	22A	2A	23A,40B	5	75	"	15,16,18
P3	"	22A	2A	3A,18B	10	150	"	15,17,18
P4	"	2A	22A	3A,18B	10	150	"	"
P5	"	7A,11A,16A, 19A,20A, 31A,40A,24A	6A	12A,2A	6	50	"	"
P6	"	6A,11A,16A, 19A,20A, 31A,40A,24A	7A	"	"	"	"	"
P7	"	6A,7A,16A, 19A,20A, 31A,40A,24A	11A	"	"	"	"	"
P8	"	6A,7A,11A, 19A,20A, 31A,40A,24A	16A	"	"	"	"	"
P9	"	6A,7A,11A, 16A,20A, 31A,40A,24A	19A	"	"	"	"	"
P10	"	6A,7A,11A, 16A,19A, 31A,40A,24A	20A	"	"	"	"	"
P11	"	6A,7A,11A, 16A,19A, 20A,40A,24A	31A	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P12	Propagation delay time (t_{PLH}), C_L = 50 pF	6A,7A,11A, 16A,19A, 20A,31A,24A	40A	12A,2A	6	50	ns	15,17,18
P13	"	23B,25B, 31B,34B, 36B,3B,9B	22B	29B,22A	"	"	"	"
P14	"	22B,25B, 31B,34B, 36B,3B,9B	23B	"	"	"	"	"
P15	"	22B,23B, 31B,34B, 36B,3B,9B	25B	"	"	"	"	"
P16	"	22B,23B, 25B,34B, 36B,3B,9B	31B	"	"	"	"	"
P17	"	22B,23B, 25B,31B, 36B,3B,9B	34B	"	"	"	"	"
P18	"	22B,23B, 25B,31B, 34B,3B,9B	36B	"	"	"	"	"
P19	"	22B,23B, 25B,31B, 34B,36B,9B	3B	"	"	"	"	"
P20	"	22B,23B, 25B,31B, 34B,36B,3B	9B	"	"	"	"	"
P21	"	37B	18B	17B	5	25	"	15,16,18
P22	"	18B	37B	17B	"	"	"	"
P23	"	39B	40B	20B	"	"	"	"
P24	"	40B	39B	20B	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P25	Propagation delay time (t_{PHL}), C_L = 50 pF	2A	22A	23A,40B	4	30	ns	15,16,18
P26	"	22A	2A	23A,40B	4	30	"	15,16,18
P27	"	22A	2A	3A,18B	8	60	"	15,17,18
P28	"	2A	22A	3A,18B	8	60	"	"
P29	"	7A,11A,16A, 19A,20A, 31A,40A,24A	6A	12A,2A	6	50	"	"
P30	"	6A,11A,16A, 19A,20A, 31A,40A,24A	7A	"	"	"	"	"
P31	"	6A,7A,16A, 19A,20A, 31A,40A,24A	11A	"	"	"	"	"
P32	"	6A,7A,11A, 19A,20A, 31A,40A,24A	16A	"	"	"	"	"
P33	"	6A,7A,11A, 16A,20A, 31A,40A,24A	19A	"	"	"	"	"
P34	"	6A,7A,11A, 16A,19A, 31A,40A,24A	20A	"	"	"	"	"
P35	"	6A,7A,11A, 16A,19A, 20A,40A,24A	31A	"	"	"	"	"
P36	"	6A,7A,11A, 16A,19A, 20A,31A,24A	40A	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P37	Propagation delay time (t_{PHL}), C_L = 50 pF	23B,25B, 31B,34B, 36B,3B,9B	22B	29B,22A	6	50	ns	15,17,18
P38	"	22B,25B, 31B,34B, 36B,3B,9B	23B	29B,22A	"	"	"	"
P39	"	22B,23B, 31B,34B, 36B,3B,9B	25B	29B,22A	"	"	"	"
P40	"	22B,23B, 25B,34B, 36B,3B,9B	31B	"	"	"	"	"
P41	"	22B,23B, 25B,31B, 36B,3B,9B	34B	"	"	"	"	"
P42	"	22B,23B, 25B,31B, 34B,3B,9B	36B	"	"	"	"	"
P43	"	22B,23B, 25B,31B, 34B,36B,9B	3B	"	"	"	"	"
P44	"	22B,23B, 25B,31B, 34B,36B,3B	9B	"	"	"	"	"
P45	"	37B	18B	17B	4	40	"	15,16,18
P46	"	18B	37B	17B	"	"	"	"
P47	"	39B	40B	20B	"	"	"	"
P48	"	40B	39B	20B	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

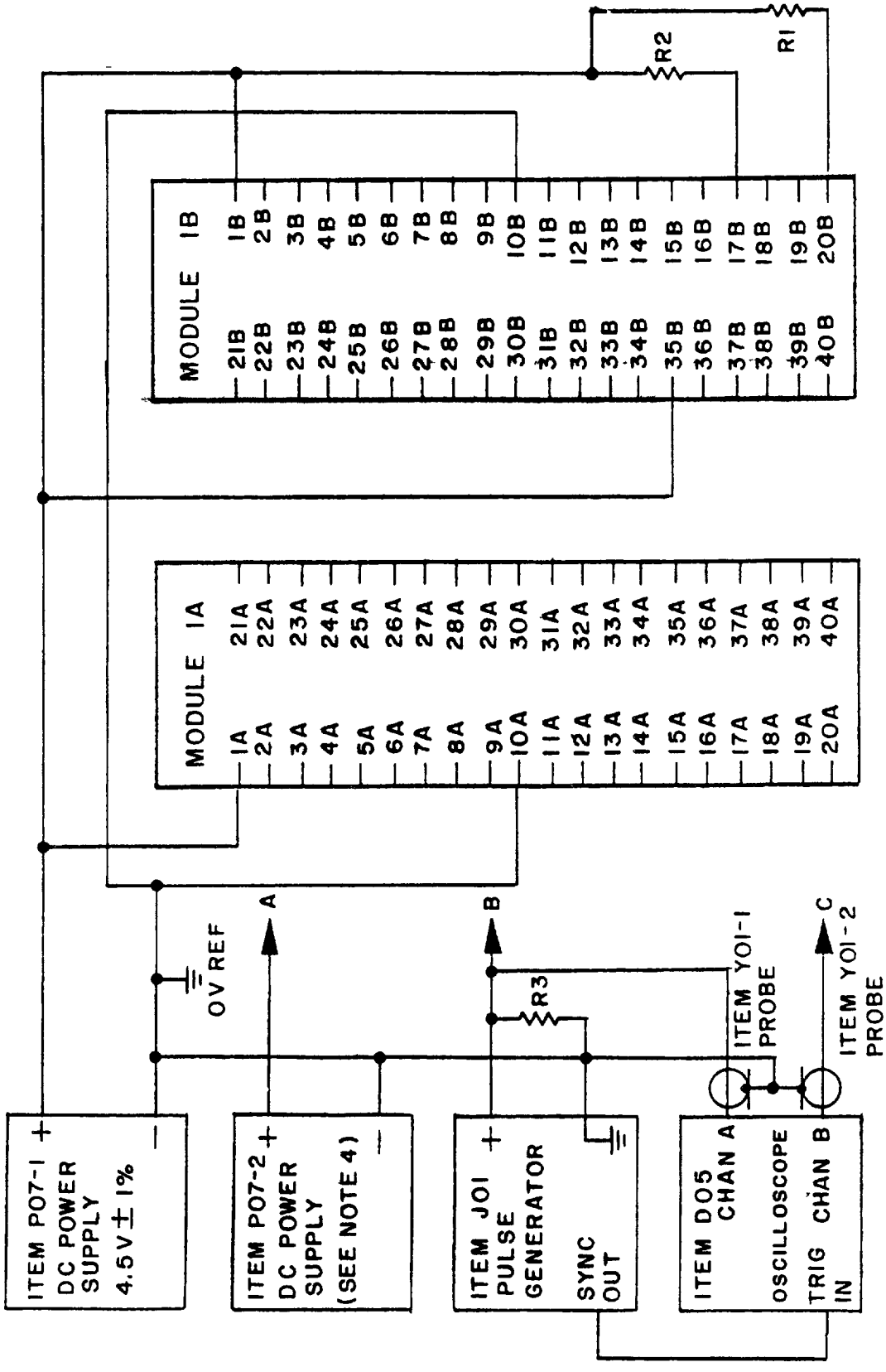


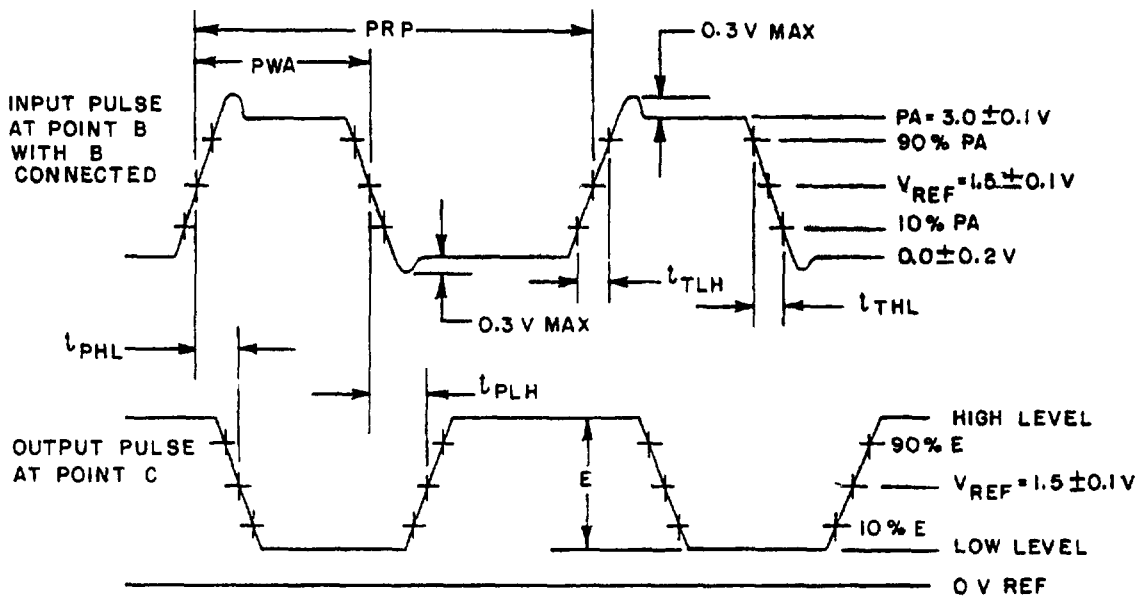
FIGURE 15. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1 = R2 = 1 kΩ ±5%, 1/8 W.
2. R3 = 51Ω ±5%, 1/2 W.
3. Load output pins 17B and 20B with the load as specified on figure 18.
4. Adjust item P07-2 as follows:
   0.7 V ±1% for test conditions P1-P20.
   2.5 V ±1% for test conditions P21-P24.

FIGURE 15. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIV - Continued.

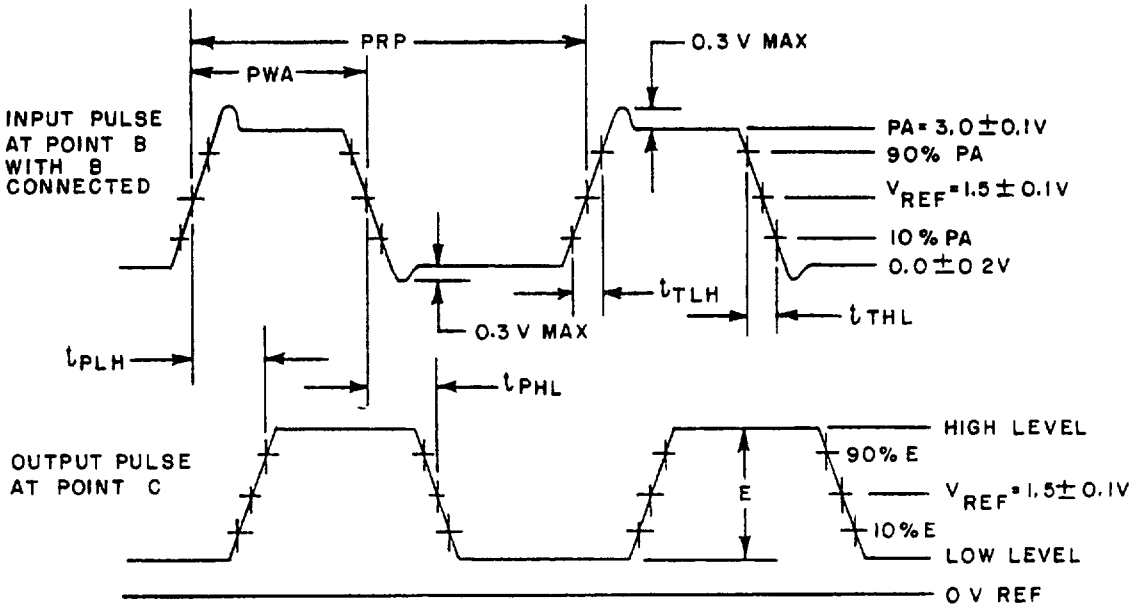
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.



NOTES.
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWA = 500 ±10 ns.
   PRP = 1,000 ±50 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$0.5 V.
3. The lead length of electrical conductors between R1 and input pins, between probes Y01-1 and Y01-2 and output pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 16. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV.

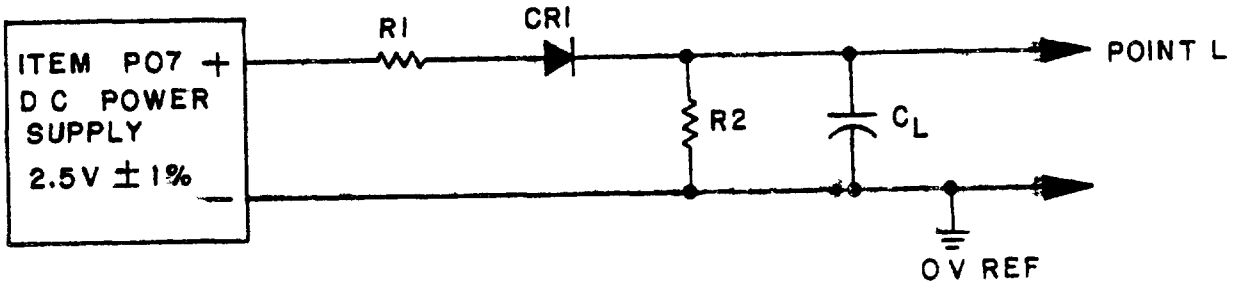
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and output pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 17. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.5 V.
3. R2 = 2.08 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q:
   C_L = 150 ±5 pF (including probe and parasitic capacitance).
5. Capacitor C_L is not required for dc electrical test conditions.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

FIGURE 18. MFG logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:55Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/344(EC)	MODULES, STANDARD ELECTRONIC ENCODER, FAULT ISOLATION, KEY CODE MFG

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. ____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:55Z
Check the source to verify that this is the current version before use.